

LLM-Powered Automated Resume Matcher & Score Evaluator

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Period of Internship: 18th May 2026 – 31st July 2026

Report submitted to:

IDEAS – Institute of Data Engineering, Analytics and Science
Foundation, ISI

Kolkata

1. Abstract

The rapid growth in job applications has made manual resume screening increasingly time-consuming, inconsistent, and prone to overlooking qualified candidates. Traditional Applicant Tracking Systems (ATS) primarily rely on keyword matching, often failing to capture the semantic meaning of candidate profiles and job requirements. This project proposes an LLM-powered Automated Resume Matcher and Score Evaluator that leverages Retrieval-Augmented Generation (RAG), semantic embeddings, and vector databases to perform intelligent and context-aware candidate evaluation.

Resume and job description datasets were preprocessed and transformed into semantic embeddings, which were indexed using ChromaDB for efficient retrieval. The retrieved resumes were analysed by a Large Language Model to evaluate skills, experience, education, and project relevance. Based on these evaluations, the system generated candidate match scores, ranked applicants, and produced explainable recommendations for recruiters. The proposed solution demonstrates how LLM-based techniques can improve recruitment efficiency, reduce manual effort, and support faster, fairer, and more informed hiring decisions.

2. Introduction

The recruitment industry has experienced a significant transformation with the rapid advancement of Artificial Intelligence (AI) and Large Language Models (LLMs). Organizations today receive thousands of job applications for a single position, making manual resume screening a time-consuming and resource-intensive process. Traditional Applicant Tracking Systems (ATS) primarily rely on keyword matching techniques, which often fail to identify candidates whose skills and experience are described using different terminology. As a result, many qualified candidates may be overlooked, while the recruitment process becomes inconsistent and susceptible to human bias.

Recent developments in Natural Language Processing (NLP), semantic search, and Large Language Models have enabled intelligent systems to understand the contextual meaning of textual information rather than relying solely on exact keyword matches. Retrieval-Augmented Generation (RAG) further enhances the capabilities of LLMs by retrieving the most relevant information from external knowledge sources before generating responses. Combined with semantic embeddings and vector databases such as ChromaDB, these technologies enable more accurate and explainable candidate evaluation.

The objective of this project, "**LLM-Powered Automated Resume Matcher and Score Evaluator**," is to develop an intelligent recruitment assistant capable of analysing resumes against job descriptions using semantic understanding. The system preprocesses resume data, generates semantic embeddings, stores them in a vector database, retrieves the most relevant resume using a RAG pipeline, and utilizes a Large Language Model to evaluate candidates. The final output includes candidate match scores, rankings, and LLM-generated recommendations, enabling recruiters to make faster and more informed hiring decisions while reducing manual effort.

This project integrates concepts from Machine Learning, Deep Learning, Natural Language Processing, Information Retrieval, and Generative AI to build a scalable and explainable resume evaluation system. It demonstrates the practical application of modern AI technologies in solving real-world recruitment challenges through automated, data-driven decision-making.

Training Topics Covered During the First Two Weeks of Internship

During the initial phase of the internship, structured training sessions were conducted to build a strong foundation in Data Science, Machine Learning, and Artificial Intelligence. Although the internship training program extended into the third week with advanced topics, the internship report template requires the topics covered during the first two weeks. The topics covered during this period included:

- Introduction to the Internship Program
- Basic Statistics for Data Science
- Data Visualization
- Introduction to GitHub and Cloud Computing
- Data Science Application Development using Streamlit
- Machine Learning – I
- Machine Learning – II
- LLM Fundamentals
- Communication Skills

The training program continued into the third week with advanced sessions on **Deep Learning, Computer Vision, Agentic AI, System View of Data Science, and Reinforcement Learning**, further strengthening the understanding of modern AI and machine learning concepts.

The knowledge gained from these training sessions provided a strong theoretical and practical foundation in statistics, machine learning, software development, version control, and Large Language Models. These concepts proved invaluable in designing and implementing the proposed **LLM-Powered Automated Resume Matcher and Score Evaluator**, enabling the

effective application of semantic search, Retrieval-Augmented Generation (RAG), and LLM-based candidate evaluation.

3. Project Objectives

The primary objectives of the LLM-Powered Automated Resume Matcher and Score Evaluator are:

- **To develop an intelligent resume screening system** that leverages Large Language Models (LLMs), semantic search, and Retrieval-Augmented Generation (RAG) to perform context-aware matching between resumes and job descriptions.
- **To automate candidate evaluation and ranking** by generating match scores based on key factors such as technical skills, work experience, educational qualifications, and project relevance, thereby reducing manual screening effort.
- **To improve the accuracy of resume matching** by utilizing semantic embeddings and vector databases (ChromaDB), enabling the system to understand contextual similarities beyond traditional keyword-based approaches.
- **To design a scalable and efficient recruitment solution** that enhances hiring consistency, reduces screening time, and supports organizations in selecting the most suitable candidates more effectively.

4. Methodology

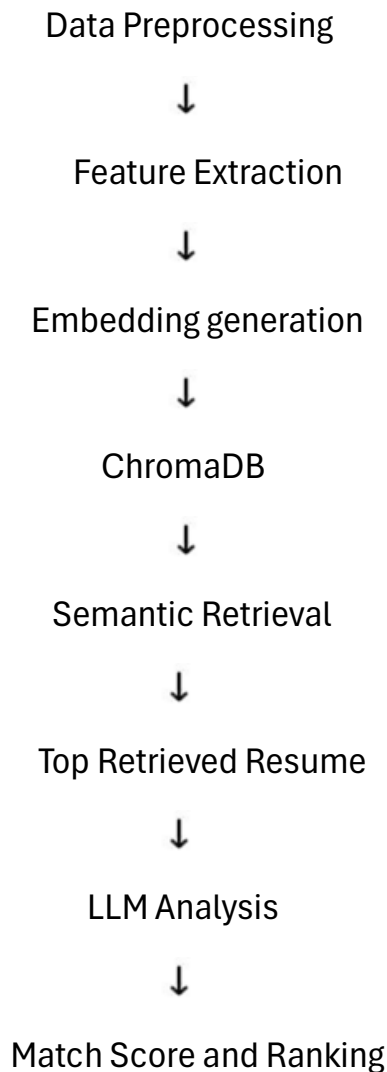
This project follows an end-to-end Retrieval-Augmented Generation (RAG) pipeline to automate resume screening. The system combines data preprocessing, semantic embeddings, vector database retrieval, and Large Language Models (LLMs) to match resumes with job descriptions. Instead of relying only on keyword matching, it understands the meaning of the text and recommends the most suitable candidates.

The system is designed to work like an Applicant Tracking System (ATS). A recruiter can select one or more job descriptions, and the system retrieves the most relevant resumes using semantic search. These resumes are then evaluated by the LLM, which generates match scores, candidate rankings, recommendations, and skill-gap analysis.

4.1 Overall Workflow

Resume Dataset + Job Description Dataset





4.2 Data Collection

Two publicly available datasets from Kaggle were used in this project.

Resume Dataset

The original resume dataset contains approximately **54,000 structured resumes**, with candidate information stored across multiple tables such as personal details, education, work experience, technical skills, and abilities. These tables were combined using a common **person_id** to create complete candidate profiles.

For the current implementation, a **representative sample of 5000 resumes** was selected after preprocessing. This reduced the computational load while allowing the resume matching pipeline to be developed and evaluated effectively.

Job Description Dataset

The job description dataset contains multiple job postings with details such as Job Title, Role, Experience, Qualifications, Skills, Responsibilities, and Job Description.

For prototype testing, **10 job descriptions** were selected from the dataset. These job descriptions were used to evaluate the semantic retrieval and LLM-based candidate scoring process.

Although the current implementation uses **5000 resumes and 10 job descriptions**, the proposed system is designed to scale to a much larger number of resumes and multiple job descriptions without changing the overall workflow.

4.3 Data Preprocessing

Before matching resumes with job descriptions, both datasets were cleaned and prepared.

The resume tables were merged using the common **person_id** so that each candidate had one complete profile.

The preprocessing steps included:

- Combining information from different resume tables.
- Handling missing values.
- Removing duplicate records.
- Converting all text to lowercase.
- Removing unnecessary symbols and special characters.
- Removing extra spaces.
- Combining education, work experience, technical skills, and abilities into a single **resume_text** field.

The job description dataset was also cleaned by selecting the required columns and standardizing the text.

After preprocessing, both resumes and job descriptions were ready for embedding generation.

4.4 Feature Extraction

Important information was extracted from both resumes and job descriptions.

From each resume, the following details were collected:

- Technical skills
- Educational qualifications
- Work experience
- Previous job roles
- Candidate abilities

From each job description, the following information was extracted:

- Required skills
- Educational qualifications
- Experience requirements
- Job responsibilities

These details were combined into text so that they could be converted into semantic embeddings.

4.5 Semantic Embedding Generation

The cleaned resumes and job descriptions were converted into numerical vectors using the **Sentence Transformer (all-MiniLM-L6-v2)** model.

Unlike traditional ATS systems that only compare keywords, semantic embeddings help the system understand the meaning of the text.

For example, the phrases "**Machine Learning**" and "**Predictive Modelling**" are understood as related concepts even though they use different words.

Resume embeddings were generated once and stored so they could be reused whenever a recruiter searched for candidates.

4.6 Vector Database and Semantic Retrieval

The resume embeddings and their related information were stored in **ChromaDB**, which acts as the system's vector database.

ChromaDB stores the embeddings and quickly retrieves resumes that are most similar to a selected job description.

When a recruiter selects one or more job descriptions, the system creates embeddings for those job descriptions and compares them with the stored resume embeddings.

The **top three** resumes with the highest semantic similarity are retrieved from ChromaDB and passed to the next stage for LLM-based evaluation.

Cosine similarity was also used to measure how closely resumes matched each job description. These similarity values were later used to create heatmaps for analysis.

4.7 RAG-Based LLM Candidate Evaluation

The project uses a **Retrieval-Augmented Generation (RAG)** approach.

Instead of sending every resume to the LLM, only the **top three semantically similar complete resumes** retrieved from ChromaDB are provided as context. Since resume embeddings are generated beforehand and stored in the vector database, only the retrieved resumes are evaluated by the LLM, reducing inference time and token usage.

The LLM evaluation was performed using the **Groq API** with the **Llama-3.1-8B-Instant** model.

Two prompts were used during evaluation:

- **System Prompt:** Instructs the LLM to behave like an unbiased recruiter and evaluate candidates only using the information available in the resume and job description.
- **User Prompt:** Provides the selected job description and retrieved resume, asking the LLM to score the candidate based on Skills, Experience, Education, Domain Relevance, Keyword Match, and Resume Quality, along with a short recommendation.

The LLM outputs individual scores for skills, experience, education, domain relevance, keyword match, and resume quality. These scores are extracted programmatically using regular expressions and combined using a weighted scoring formula to calculate the final candidate score.

4.8 Candidate Ranking and Report Generation

After the LLM evaluates the retrieved resumes, it generates scores for different aspects of each candidate, such as skills, experience, education, domain relevance, keyword match, and overall resume quality. These evaluation scores are combined to calculate a final Match Score for every candidate.

Based on the final score, candidates are ranked in descending order according to their suitability for the selected job description. For each job description, the system returns the Top-3 recommended candidates along with:

- Resume ID
- Candidate Name
- Match Score
- Rank
- Recommendation generated by the LLM

The results are compiled into an ATS screening report, making it easier for recruiters to compare candidates and shortlist the most suitable applicants.

In addition to the ranking, the report provides a recommendation generated by the LLM summarizing the candidate's suitability for the selected role, enabling recruiters to make more informed hiring decisions.

4.9 Tools and Libraries Used

Category	Tool / Technology
Programming Language	Python
Data Processing	Pandas, NumPy
Embedding Model	Sentence Transformers (all-MiniLM-L6-v2)
Vector Database	ChromaDB
Similarity Measure	Cosine Similarity (Scikit-learn)
Large Language Model	Llama-3.1-8B-Instant (via Groq API)
Environment Management	python-dotenv
Visualization	Matplotlib

4.10 Visualization

Several visualization techniques were used to analyse and interpret the performance of the proposed resume screening system.

- **Cosine Similarity Heatmap:** Visualizes the semantic similarity between candidate resumes and selected job descriptions, demonstrating the effectiveness of vector-based retrieval.
- **Candidate Ranking Bar Chart:** Displays the final match scores of shortlisted candidates for each job description, enabling recruiters to compare candidate suitability visually.

5. Data Analysis and Results

5.1 Dataset Analysis

The original resume dataset contained approximately **54,000 structured resumes**. After preprocessing and cleaning, a random sample of **5,000 resumes** was selected for embedding generation and semantic retrieval. Similarly, **10 job descriptions** were selected from the job description dataset for evaluation. This sample size was sufficient to validate the proposed resume matching pipeline while keeping the processing time manageable.

5.2 Semantic Retrieval Analysis (Cosine Similarity Heatmaps)

Semantic embeddings were generated using the **Sentence Transformer (all-MiniLM-L6-v2)** model. Cosine similarity was then computed between each selected job description and all **5,000 resume embeddings** to identify semantically similar candidates.

The similarity scores were visualized using heatmaps, where each row represents a resume and the colour intensity indicates the similarity score. The heatmaps show that only a small group of resumes achieved high similarity scores for each job description, demonstrating that the semantic retrieval process effectively identifies relevant candidates. The top matching resumes were retrieved from **ChromaDB** and passed to the LLM for detailed evaluation and ranking.

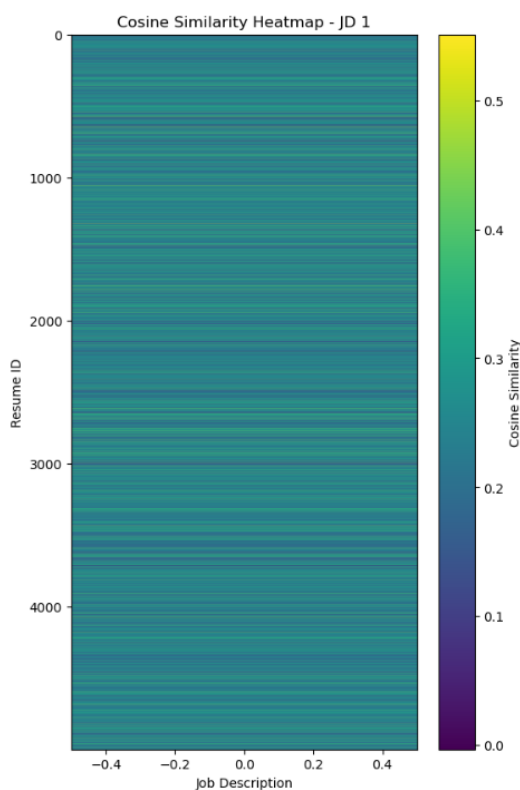


Fig 5.1 : Cosine Similarity Heatmap for

Job Description 1

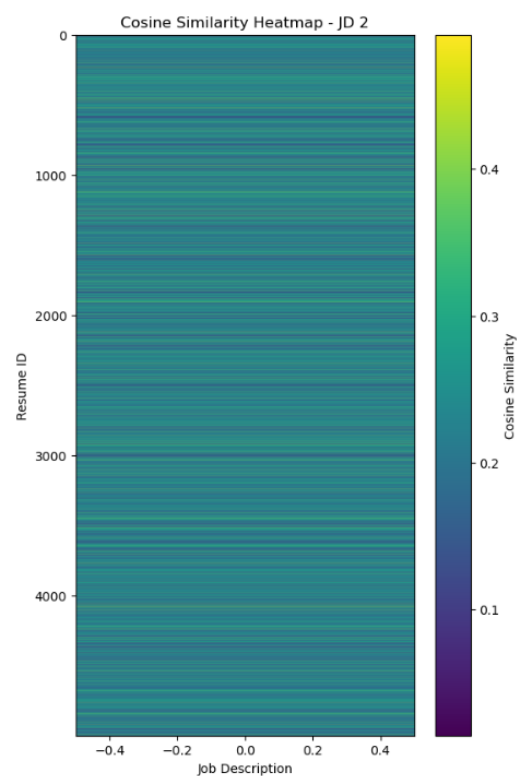


Fig 5.2 : Cosine Similarity Heatmap for

Job Description 2

5.3 LLM-Based Candidate Ranking

For each selected job description, the top-3 resumes retrieved from ChromaDB were evaluated using the Llama-3.1-8B-Instant model through the Groq API. The LLM assessed each resume based on skills, experience, education, domain relevance, keyword match, and resume quality, and generated a Final Match Score (0–100) along with a recommendation.

The candidates were then ranked according to their match scores. The bar charts below show the rankings for sample job descriptions, highlighting the most suitable candidates identified by the system.

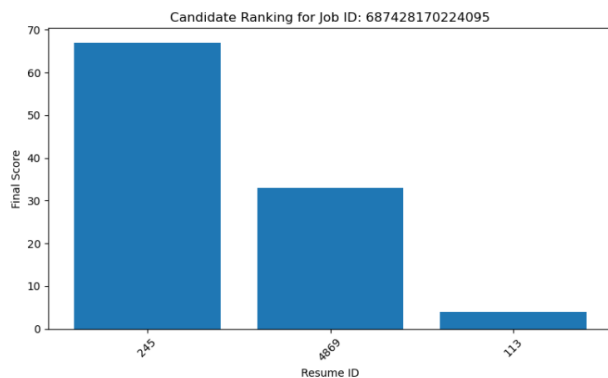


Fig 5.3: Candidate Ranking for Job Description 1

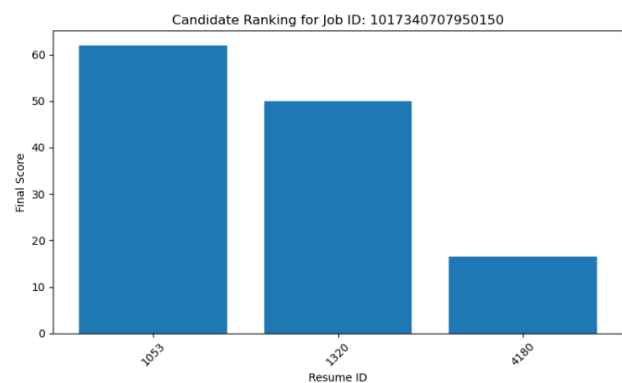


Fig 5.4: Candidate Ranking for Job Description 2

5.4 ATS Screening Report

The final output of the system is an ATS Screening Report (`ats_report.csv`). For each job description, the report lists the shortlisted candidates along with their Resume ID, Rank, category-wise scores, Final Match Score, and the LLM-generated recommendation.

Job ID	Job Title	Rank	Resume ID	Skills	Experience	Education	Domain	Keyword	Resume Quality	Final Score	Recommendation
687428170224095	Account Manager	1	245	20	20	0	10	5	8	67.0	The candidate has a strong background in projec
687428170224095	Account Manager	2	4869	10	5	0	5	5	8	33.0	The candidate has some relevant skills such as cu
687428170224095	Account Manager	3	113	0	0	0	0	0	8	4.0	The candidate has a strong resume with a clear a
1017340707950150	Procurement Manager	1	1053	20	15	0	10	5	8	62.0	The candidate has relevant experience in procure
1017340707950150	Procurement Manager	2	1320	15	10	0	10	5	8	50.0	The candidate has relevant skills such as strategic
1017340707950150	Procurement Manager	3	4180	10	0	0	0	0	5	16.5	The candidate's resume does not clearly match th
1690092345385690	Business Development Manager	1	113	15	5	10	5	5	8	46.67	The candidate has a strong educational backgrou
1690092345385690	Business Development Manager	2	4869	15	5	10	5	5	8	46.67	The candidate has relevant education and some s
1690092345385690	Business Development Manager	3	1320	10	5	0	5	5	8	33.0	The candidate has some relevant skills, such as s

Fig 5.5 : ATS Report with Candidate Scores and Rankings

6. Conclusion

This project successfully developed an LLM-Powered Automated Resume Matcher & Score Evaluator using a Retrieval-Augmented Generation (RAG) approach. The system combines data preprocessing, semantic embeddings, ChromaDB-based retrieval, and LLM evaluation to identify and rank suitable candidates for a given job description. The pipeline was validated using 5,000 resumes and 10 job descriptions, demonstrating the system's ability to retrieve semantically relevant candidates beyond simple keyword matching. By generating match scores and recommendations, the system provides recruiters with an efficient and explainable

candidate screening process, reducing manual effort while improving the accuracy and consistency of resume evaluation.

Future Enhancements

- **Real-Time Resume Upload and Evaluation:** Allow recruiters to upload new resumes in PDF or DOCX format. The system can automatically process the uploaded resumes, generate embeddings, and evaluate them against selected job descriptions without requiring them to be part of the existing resume database.
- **Interactive Recruiter Dashboard:** Develop a user-friendly web interface that allows recruiters to submit multiple job descriptions, view the top-ranked candidates for each role, access AI-generated recommendations, and download ATS reports for further screening.
- **Advanced LLM Evaluation:** Explore larger or fine-tuned Large Language Models to improve the accuracy, consistency, and explainability of candidate scoring and recommendations.

7. Appendices

Appendix A: References

- <https://www.w3schools.com/>
- <https://developers.openai.com/>
- <https://www.geeksforgeeks.org/nlp/sentence-transformer/>

Appendix B: Source Code Repository

GitHub Link - <https://github.com/vivekvardhanreddykalvaug2025-ui/llm-resume-matcher>